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LEARNER GUIDE

For



Generative AI for Project Management

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Introduction

This Learner Guide accompanies the course Generative AI for Project Management (C013), conducted by Tertiary Infotech Academy Pte Ltd. It provides detailed step-by-step instructions for all 25 hands-on activities, organised by the three course topics and mapped to the three learning outcomes.

Every activity in this course is GenAI-first. You will prompt a Generative AI tool to produce a first-pass project artifact — a charter, a work breakdown structure, a risk register, a schedule — and then critically review, correct and complete it. That review step is the skill this course teaches. GenAI drafts; you decide.

Use this guide alongside the course slides. The slides show each activity's goal, the prompt and the artifact you produce; this guide carries the full step-by-step detail you follow at your own pace.

Course Learning Outcomes

- LO1: Develop project parameters with GenAI according to business requirements.
- LO2: Resolve project delivery challenges by optimising resource and project budgets with GenAI.
- LO3: Anticipate and devise risk mitigation measures with GenAI to ensure project success.

Skills Framework Alignment

TSC Abilities

TSC Knowledge

Before You Start — Your GenAI Toolkit

What you need

- A laptop with a modern web browser and internet access.
- An account on at least one Generative AI tool — ChatGPT (chat.openai.com), Claude (claude.ai), Google Gemini (gemini.google.com) or Microsoft Copilot (copilot.microsoft.com). A free account is sufficient.
- The Project Manager Buddy GPT, used throughout the activities: <https://chatgpt.com/g/g-hUcEmp6Rp-project-manager-buddy-project-management>
- The course template pack (Excel, Word and PowerPoint templates used in the activities): <https://drive.google.com/drive/folders/1-FY7C8j7NfxilnKwSgK2c6hgTRT-8K73?usp=sharing>
- Microsoft Excel, Word and PowerPoint (or Google Sheets, Docs and Slides) to complete the templates.

- No installation is required for any of the five course ed-tools — they all run in the browser.

Set up before Activity 1

- Sign in to your chosen GenAI tool and confirm you can start a new chat.
- Open the Project Manager Buddy GPT in a second browser tab and keep it open all course.
- Download the template pack and unzip it to a folder you can find quickly.
- Have a real project of your own in mind — one you know well enough to judge whether AI output is sensible.

Working safely with GenAI at work

Everything you put into a public GenAI tool may be used to improve that model. Before you use these techniques on real work, confirm what your organisation permits.

- Never paste client data, personal data, credentials or commercially sensitive information into a public tool.
- Your organisation may mandate an enterprise tool (such as Microsoft Copilot with commercial data protection) instead.
- Anonymise a scenario before prompting: replace real names, figures and identifiers with placeholders.
- Treat every figure, date, citation and source the model produces as unverified until you check it.
- Be transparent where GenAI materially shaped a deliverable you are handing to a client or sponsor.
- Keep the human in the loop — approval decisions stay with people, not models.

Conventions used in every activity

- Prompts you type into a GenAI tool are shown in a monospaced block.
- Placeholders such as [your project], [Industry] and [Project type] are replaced with your own details.
- Most activities are done in a group of 3–5; one person creates the shared ed-tool board and shares the link.
- Each activity ends with a 'Check your work' test — use it to confirm you have produced what was asked.
- Where an ed-tool cannot export a file, take a screenshot as your evidence.

Reference — The Ten Prompting Patterns

These ten patterns are taught in Topic 1 and used throughout the course. Keep this section open during the activities; the pattern name tells you which shape of prompt to reach for.

The anatomy of a good prompt

- **ROLE** — who the AI should act as ('Act as an expert project manager in the construction industry').
- **CONTEXT** — your project, industry, constraints and what you already know.
- **TASK** — exactly what to produce, and how much of it.
- **FORMAT** — a table, a list, a document; the columns you want; the length.
- **CHECK** — instruct it to mark assumptions and cite sources so you can verify.

Pattern — Persona

Frame the output from the perspective of a specific project role — or a non-human entity such as project management software.

Example prompt

Act as a stakeholder of the project. Perform an assessment of the project charter and provide a list of recommendations to improve scope clarity and completeness.

Pattern — Question Refinement

Ask GenAI to improve your question when there is no clarity about the problem you are trying to solve.

Example prompt

Within the scope of creating a cost-benefit analysis of outsourcing (total cost: US\$60,000) versus in-house hiring (total cost: US\$70,000), suggest a better version of the question to be used.

Pattern — Flipped Interaction

Let GenAI interview you — it asks the questions, one at a time, to find the gaps in your thinking.

Example prompt

I would like you to ask questions to help with my preparation for a project update with my stakeholders. Ask me questions one at a time so you can build the next question based on my responses.

Pattern — Alternative Approaches

Generate several options, then compare their pros and cons before you decide.

Example prompt

I am a construction project manager experiencing a delay in completion of some tasks. I want to identify the root cause(s) of the problem. Suggest at least two options to deal with the problem(s) identified and compare the pros and cons of each. Present the results in a table.

Pattern — Cognitive Verifier

Force GenAI to break a complex question into smaller sub-questions and then combine the answers.

Example prompt

A software development project has a US\$100,000 budget and a timeline of six months. After three months the actual cost is US\$60,000 and the project is 50% complete. What are the risks of not completing on time and on budget? Am I over or under budget? Am I behind or ahead of schedule? What is the earned value and estimate to complete? Combine the answers to produce the final answer to the first question.

Pattern — Fact Check List

For outputs that rest on external information, require the facts to be listed separately with links to sources.

Example prompt

Generate a list of the 10 most common frequent-flier challenges in the airline industry based on facts. The set of facts should be inserted at a specific point in the output and correctly show links to sources.

Pattern — Visualisation Generator

Have GenAI write the prompt for a text-to-image or diagram tool so you can visualise project information.

Example prompt

I want to generate a prompt that I can provide to a diagram tool to visualise the 10 most common sources of project scope creep for my project kickoff presentation.

Pattern — Reflection

Ask for the reasoning and assumptions behind a recommendation — essential for risk and people decisions.

Example prompt

I need to provide feedback to a project team member who made a mistake with financial implications. Explain the reasoning and assumptions behind your recommendation on how to approach this meeting with them.

Pattern — Risk Matrix Template

A long-form template that produces a complete, industry-specific risk matrix in one pass.

Example prompt

Imagine you are an expert project manager in the [Industry], focusing on [Project type] projects. Develop a comprehensive risk matrix. For each risk include: Risk name, Description, Level of impact (high/medium/low) and Likelihood (high/medium/low). Cover budget, schedule, technology, regulatory, resource, scope creep, stakeholder, data security, quality and external dependency categories. Output as a table.

Pattern — Decision Making Template

Decompose a complex decision into smaller questions that together answer it.

Example prompt

Imagine you are an expert project manager in the [Industry] working on a [Project type] project. Break the complex question "[Complex question?]" into smaller questions that collectively answer it. Consider the range of outcomes, sources of resistance, risks and mitigation plans, and key milestones. Provide one-to-three-sentence answers to each, then ask me where I would like to go deeper.

Reference — The Five Course Ed-Tools

Five browser-based collaborative tools are used in the activities. All five are free, need no installation and no account: one person creates the workspace and shares a link or QR code, and everyone else joins by typing a display name.

RACI Collaborative Matrix

Drag A / R / C markers into a live responsibility matrix with your team.

URL: <https://alfredang.github.io/raci/>

Business Model Canvas Studio

The nine-block Osterwalder canvas — validate the business model behind your project.

URL: <https://alfredang.github.io/bcm/>

Design Thinking Studio

Empathise, Define, Ideate, Prototype, Test — with dot-voting on ideas.

URL: <https://alfredang.github.io/designthinking/>

Scrum Agile Transformation Studio

Sprint planning, planning poker, burndown and retrospectives; exports JSON/CSV.

URL: <https://alfredang.github.io/scrum/>

Digital AI Transformation Roadmap Studio

Initiatives, impact-vs-effort matrix, 12-month timeline, risk register and KPIs.

URL: <https://alfredang.github.io/digitaltransformation/>

Important practical notes

- Only the Scrum Agile Transformation Studio exports a file (JSON, CSV or a printable HTML report). For the other four tools, capture your result with a screenshot.
- There is no login and no per-user account — you are identified only by the display name you type when joining.
- The RACI tool provides A, R and C markers only; there is no 'Informed' marker, so record informed parties in your communications plan instead.
- The RACI board arrives pre-seeded with a sample GenAI project matrix — rename the roles and activities to match your own project rather than starting from an empty grid.
- The Business Model Canvas, Design Thinking and Digital Transformation tools all start empty, so budget time for data entry.

Topic 01 — Project Initiation and Planning with GenAI (LO1 • K2, A1)

Charter · business case · stakeholders · scope · WBS · schedule · cost

Key concepts

- What a project is — A unique product, service or result; time-limited; drives change and enables value creation.
- GenAI as your planning co-pilot — ChatGPT, Claude, Gemini and Copilot draft charters, WBS, schedules and estimates in minutes — you stay accountable for the judgement.
- Ten prompting patterns — Persona, Question Refinement, Flipped Interaction, Alternative Approaches, Cognitive Verifier, Fact Check List, Visualisation Generator, Reflection and the risk/decision templates.
- Project Charter — Authorises the project, names the sponsor and PM, and fixes objectives, budget and success criteria.
- Business case & value — Cost-benefit analysis and a Business Model Canvas prove economic feasibility before work starts.
- Scope & requirements — Collect requirements, build the Requirements Traceability Matrix, write the scope statement, decompose into a WBS.
- Schedule — Define, sequence and estimate activities; build the network diagram and find the critical path.
- Cost — Analogous, parametric, bottom-up and three-point estimating; the cost baseline plus contingency and management reserves.

Activity 1 — Prompting Patterns for Project Managers

Learning outcome: LO1 — Apply structured prompting patterns to produce usable project management output.

Goal: Work through the ten prompting patterns on a project of your own. You will see the same question answered badly by a vague prompt and well by a structured one, and you will keep a personal prompt library for the rest of the course.

What you'll produce

A personal prompt library of 10 tested prompts, each with the output it produced and your critique. (Tools: ChatGPT · Claude · Gemini · Copilot · Project Manager Buddy GPT.)

Activity package

Open [activities/activity-01-prompting-patterns-for-project-managers/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Act as a stakeholder of the project. Perform an assessment of the project charter and provide a list of recommendations to improve scope clarity and completeness.

Step-by-step

1. Open your preferred GenAI tool (ChatGPT, Claude, Gemini or Copilot) and sign in. Open the Project Manager Buddy GPT in a second tab: <https://chatgpt.com/g/g-hUcEmp6Rp-project-manager-buddy-project-management>
2. Choose a real project from your own workplace — one you know well enough to judge whether the AI output is sensible. Write down its objective, budget range and timeline in one sentence each.
3. Start with a deliberately vague prompt: 'Help me with my project.' Note how generic the answer is. This is your baseline.
4. Now apply the Persona pattern: 'Act as a stakeholder of the project. Perform an assessment of the project charter and provide a list of recommendations to improve scope clarity and completeness.' Compare against the baseline.
5. Apply the Flipped Interaction pattern so the AI interviews you: 'Ask me questions one at a time to prepare for a stakeholder update on [your project].' Answer at least five questions and note which gap it exposed.
6. Apply the Cognitive Verifier pattern to a complex question about your project — one that needs several sub-answers combined. Observe how it decomposes the question.

7. Apply the Fact Check List pattern to a question that depends on external industry data. Verify at least two of the sources it cites actually exist and say what it claims.
8. Record each prompt, the output and a one-line critique in your prompt library. Mark any output that was confidently wrong — this is the habit that protects you all course.

 Check your work

You have 10 prompts in your library, each with output and a critique, and you can state one concrete case where GenAI produced a plausible but wrong answer that you caught.

Activity 2 — Business Case and Cost-Benefit Analysis with GenAI

Learning outcome: LO1 — Establish the economic feasibility of a project using GenAI-assisted analysis.

Goal: Before a project is authorised somebody has to prove it is worth doing. You will use GenAI to draft a cost-benefit analysis and a one-page business case, then validate the business model behind it on the Business Model Canvas Studio.

What you'll produce

A completed cost-benefit analysis, a one-page business case, and a validated 9-block Business Model Canvas. (Tools: GenAI tool · Business Model Canvas Studio (alfredang.github.io/bcm) · Cost Benefit Analysis.xlsx · Business Case on a Page.docx.)

Activity package

Open [activities/activity-02-business-case-and-cost-benefit-analysis-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

List the one-off and recurring costs, and the quantified annual benefits, of [your project] over 3 years. Present the result as a table with a column explaining how each figure will be verified.

Step-by-step

1. Form a group of 3–5. Download 'Cost Benefit Analysis.xlsx' and 'Business Case on a Page.docx' from the course template pack.
2. Agree on a project scenario for your group — for example introducing a GenAI knowledge assistant for your customer service team.
3. Prompt GenAI: 'List the one-off and recurring costs, and the quantified annual benefits, of [your project] over 3 years. Present as a table with a column for how each figure would be verified.'

4. Challenge the output with the Fact Check List pattern — ask it to separate figures it knows from figures it assumed. Replace every assumed figure with your own estimate.
5. Enter the validated figures into Cost Benefit Analysis.xlsx and read off the payback period and net benefit.
6. Open the Business Model Canvas Studio at <https://alfredang.github.io/bcm/> — one member clicks 'Create New Canvas', enters a name and canvas title, then shares the QR code or invite link so the rest of the group joins.
7. Prompt GenAI: 'Draft a Business Model Canvas for [your project]. Give me 2–3 sticky notes for each of the nine blocks: Key Partners, Key Activities, Key Resources, Value Propositions, Customer Relationships, Channels, Customer Segments, Cost Structure, Revenue Streams.'
8. Add the AI's suggestions as sticky notes in the matching blocks. Set each note's status to Draft. Use the (i) 'Show guidance' button on any block you are unsure about.
9. As a group, review every note. Change the status to Validate for notes you have evidence for and Confirmed for notes you are certain of. Delete anything the AI invented that you cannot support.
10. Complete Business Case on a Page.docx using your validated numbers and canvas, and share your findings with the class.

 Check your work

Your canvas has all nine blocks populated, every note carries a Draft/Validate/Confirmed status, and your business case states a payback period you can defend with your own figures.

Activity 3 — Develop the Project Charter with GenAI

Learning outcome: LO1 — Develop project parameters with GenAI according to business requirements.

Goal: The charter authorises the project and gives the PM the right to apply resources. You will generate a full charter with GenAI and then correct it against what a charter must actually contain.

What you'll produce

An approved-quality Project Charter for your group's project. (Tools: GenAI tool · Project Manager Buddy GPT · Project Charter Template.xlsx.)

Activity package

Open activities/activity-03-develop-the-project-charter-with-genai/. It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Write a project charter for launching an internal knowledge-sharing platform. Include objectives, scope, risks, stakeholders and success criteria. Mark every assumption clearly.

Step-by-step

1. Stay in your group of 3–5 and open 'Project Charter Template.xlsx' from the template pack.
2. Prompt GenAI: 'Write a project charter for launching an internal knowledge-sharing platform. Include objectives, scope, risks, stakeholders, and success criteria.' — or substitute your own project.
3. Check the draft against the required charter contents: sponsor, project manager and key stakeholders named; project description with measurable objectives; business need and financial goals; summary milestone schedule; assumptions, constraints and boundaries; approval requirements and approved budget; success and exit criteria.
4. List every element the AI omitted or left vague. In practice it will usually invent a budget and skip exit criteria.
5. Prompt it again to fix precisely those gaps: 'Revise the charter. Add measurable success criteria, explicit exit criteria, and mark every figure you assumed with [ASSUMPTION].'
6. Replace each [ASSUMPTION] with your group's own figure, then fill in Project Charter Template.xlsx.
7. Use the Persona pattern to stress-test it: 'Act as the project sponsor. Review this charter and list the three questions you would ask before signing it.' Answer those three questions in your charter.
8. Share your charter with the class and state which part GenAI got most wrong.

Check your work

Your charter names a sponsor and PM, has measurable objectives and explicit success AND exit criteria, and contains no unmarked AI-invented figures.

Activity 4 — Identify and Map Stakeholders with GenAI

Learning outcome: LO1 — Identify stakeholders and plan engagement using GenAI-assisted analysis.

Goal: Projects fail on stakeholders more often than on technology. You will use GenAI to generate a stakeholder register, then map power against interest and decide an engagement strategy for each.

What you'll produce

A stakeholder register, a power/interest map and a Stakeholder Engagement Assessment Matrix. (Tools: GenAI tool · Stakeholder Impact Over Influence Matrix.xlsx · Saliency Model Template.pptx · Stakeholder Engagement Assessment Matrix.xlsx.)

Activity package

Open [activities/activity-04-identify-and-map-stakeholders-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Identify potential stakeholders and their influence and interest levels for a new city-wide bike-sharing system. Explain the evidence needed to validate each rating.

Step-by-step

1. In your group, open 'Stakeholder Impact Over Influence Matrix.xlsx' from the template pack.
2. Prompt GenAI: 'Identify potential stakeholders and their influence/interest levels for a new city-wide bike-sharing system.' — or use your own project.
3. Add the stakeholders GenAI missed. It reliably finds obvious internal roles and misses regulators, unions, adjacent departments and the people who lose something if the project succeeds.
4. Plot each stakeholder on the matrix: X-axis = Influence, Y-axis = Interest. Agree the four quadrant strategies — manage closely, keep satisfied, keep informed, monitor.
5. Now apply the Saliency Model. Prompt: 'Assess each stakeholder for power, legitimacy and urgency, and classify them as definitive, dominant, dangerous, dependent, discretionary, demanding or dormant.'
6. Place each stakeholder into the correct saliency category in Saliency Model Template.pptx and reconcile any disagreement with your power/interest map.
7. Open 'Stakeholder Engagement Assessment Matrix.xlsx'. Prompt GenAI: 'List stakeholders for this project and recommend their current and desired engagement levels (Unaware, Resistant, Neutral, Supportive, Leading) with a matching engagement strategy for each.'
8. Mark C for current and D for desired against each stakeholder, and write the specific action that closes each gap. Share with the class.

Check your work

Every stakeholder has a quadrant, a salience category, a current and desired engagement level, and one named action to close the gap.

Activity 5 — Assign Responsibilities with the RACI Ed-Tool

Learning outcome: LO1 — Allocate project responsibilities unambiguously using a responsibility assignment matrix.

Goal: A RACI matrix ends the 'I thought you were doing it' conversation. You will build one collaboratively in a live browser tool, using GenAI to propose the first pass.

What you'll produce

A completed RACI matrix for your project, agreed by the whole group. (Tools: RACI Collaborative Matrix (alfredang.github.io/raci/) · GenAI tool.)

Activity package

Open [activities/activity-05-assign-responsibilities-with-the-raci-ed-tool/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

For a project with these activities [paste] and these roles [paste], propose a RACI assignment. Give exactly one Accountable role per activity and explain any ambiguous accountability.

Step-by-step

1. One member of your group opens <https://alfredang.github.io/raci/> and clicks 'Create new board'. Share the QR code or board URL from the 'Share' button so everyone joins.
2. Note the board arrives pre-seeded with a GenAI project example — roles Solution Architect, Platform Team, Security & Compliance, Data/Knowledge Owner and five activities. Rename these to match your own project by clicking each header.
3. Use '+ Add role' and '+ Add activity' to complete your grid. Aim for 4–6 roles and 6–8 activities.
4. Prompt GenAI: 'For a project with these activities [paste] and these roles [paste], propose a RACI assignment. Give exactly one Accountable per activity and explain any activity where accountability is ambiguous.'
5. Drag the A, R and C markers into the cells to match the AI's proposal. Note this tool uses A/R/C only — there is no Informed marker, so record 'informed' parties in your communications plan instead.

6. Now audit it as a group. Check the two rules that matter: exactly ONE Accountable per activity, and no role that is Accountable for everything. Click a marker to remove it, or drag it to the trash.
7. Find at least one place where the AI's suggestion was wrong for your organisation and fix it. Ask why it got it wrong — usually it assumes a generic org structure.
8. Screenshot the finished board (the tool has no export) and share it with the class.

 Check your work

Every activity row has exactly one A, at least one R, and your group can justify each assignment; you have a screenshot saved as evidence.

Activity 6 — Collect Requirements with GenAI and Design Thinking

Learning outcome: LO1 — Elicit, structure and prioritise requirements using GenAI-supported techniques.

Goal: Requirements gathering is where GenAI earns its keep. You will run brainstorming, affinity clustering and prioritisation — using the Design Thinking Studio to structure the work and dot-vote as a group.

What you'll produce

A clustered, prioritised requirements set with a MoSCoW classification. (Tools: Design Thinking Studio (alfredang.github.io/designthinking/) · GenAI tool · MoSCoW with Sliders Template.xlsx.)

Activity package

Open [activities/activity-06-collect-requirements-with-genai-and-design-thinking/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Create three user personas and their top pain points for [your project's users]. Then list 15 to 20 ideas for solving the highest-priority pain point. Mark assumptions that require user research.

Step-by-step

1. One member opens <https://alfredang.github.io/designthinking/> and clicks 'Create New Workspace'. Everyone joins with the workspace code or QR, entering their display name.
2. Go to stage 1, Empathize. Prompt GenAI: 'Create 3 user personas and their top pain points for [your project's users].' Add each persona as a note in the User Personas section and each pain point under User Pain Points.

3. Fill the Empathy Map quadrants — Says, Thinks, Does, Feels — for your primary persona, using GenAI to draft and your own judgement to correct.
4. Move to stage 2, Define. Prompt: 'Write a clear problem statement and five How Might We questions for [the pain points above].' Add these to the Problem Statement and How Might We Questions sections.
5. Move to stage 3, Ideate. Prompt: 'Provide a list of brainstorming prompts or questions related to [topic or problem] to stimulate creative thinking and generate a diverse range of ideas', then 'List 15-20 ideas for [your problem].'
6. Add the ideas to the Brainstorming Board. Then cluster them: group similar ideas and label each cluster with a clear theme — this is your affinity analysis.
7. Use the Voting & Prioritisation section, which has voting enabled. Each learner votes on the ideas they believe should be built. Read off the top five by vote count.
8. Open 'MoSCoW with Sliders Template.xlsx'. Prompt GenAI: 'Classify these requirements as Must have, Should have, Could have or Won't have for a first release, and justify each classification.'
9. Override any classification your group disagrees with — GenAI systematically over-classifies things as 'Must have'. Share your prioritised set with the class.

Check your work

Your workspace has personas, an empathy map, a problem statement, a clustered idea board with vote counts, and a MoSCoW list where you changed at least one AI classification.

Activity 7 — Build the Requirements Traceability Matrix with GenAI

Learning outcome: LO1 — Trace every requirement from source through to test and deliverable.

Goal: An RTM proves that what you built is what was asked for. GenAI can populate one quickly, but it will happily invent requirement sources — which is exactly the failure mode an RTM exists to prevent.

What you'll produce

A complete Requirements Traceability Matrix with verified sources. (Tools: GenAI tool · Requirements Traceability Matrix.xlsx.)

Activity package

Open [activities/activity-07-build-the-requirements-traceability-matrix-with-gena/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Generate 5 to 10 sample requirements for a smart-city parking app. Include Requirement ID, Description, Source, Acceptance Criterion and Test Case ID. Flag any requirement that is not testable.

Step-by-step

1. In your group, open 'Requirements Traceability Matrix.xlsx' from the template pack.
2. Prompt GenAI: 'Generate 5-10 sample requirements for a smart city parking app. Include: Requirement ID, Description, Source, and Test Case ID.' — or use your own project.
3. Try the worked example too: 'Generate a Requirements Traceability Matrix for an e-health app for appointment booking. Include requirement ID, description, stakeholder source, and test case.' Compare the structure of the two outputs.
4. Paste the output into the template. Now audit the Source column — for each row, ask 'which real stakeholder or document did this come from?'
5. Delete or re-source every requirement whose source the AI invented. This is the single most important step in this activity.
6. Link each requirement forward to the scope statement, an acceptance criterion and a test case ID so the trace is complete end to end.
7. Prompt GenAI to check your work: 'Review this RTM. Which requirements have no test case, and which test cases do not trace back to a requirement?'
8. Fix the gaps it finds and share your matrix with the class.

Check your work

Every row traces from a real, named source through to an acceptance criterion and a test case, with no orphan requirements or orphan tests.

Activity 8 — Decompose Scope into a WBS and WBS Dictionary with GenAI

Learning outcome: LO1 — Decompose project scope into work packages with a supporting dictionary.

Goal: The WBS is the backbone of the plan — everything downstream, schedule and cost, is built from it. GenAI decomposes work quickly; your job is to check the decomposition is complete and mutually exclusive.

What you'll produce

A three-level WBS and a WBS Dictionary for your work packages. (Tools: GenAI tool · WBS Template.pptx · WBS Dictionary.xlsx.)

Activity package

Open activities/activity-08-decompose-scope-into-a-wbs-and-wbs-dictionary-with-g/. It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Create a three-level work breakdown structure for implementing an internal knowledge portal for a consulting firm. Make every lowest-level item a deliverable-based work package.

Step-by-step

1. In your group, prompt GenAI: 'Create a WBS for implementing an internal knowledge portal for a consulting firm.' — or use your own project scenario.
2. Organise the output into levels 1, 2 and 3 in 'WBS Template.pptx'. Level 3 items should be work packages — small enough to assign to one person or team.
3. Apply the 100% rule: the children of any node must add up to all of that node's work, with no overlap. Ask GenAI: 'Does this WBS satisfy the 100% rule? Identify any missing or overlapping work.'
4. Check for the work GenAI habitually forgets: project management effort itself, testing, training, documentation, handover and warranty support. Add what is missing.
5. Open 'WBS Dictionary.xlsx'. Prompt GenAI: 'For each work package, produce a WBS dictionary entry with WBS code, description of work, responsible role, resources required, cost estimate, quality requirements and acceptance criteria.'
6. Review each entry — particularly the acceptance criteria, which must be testable. Rewrite any that are vague ('works well' is not an acceptance criterion).
7. Confirm the flow is intact: Scope Statement → WBS → Work Packages → Activity List, ready for scheduling. Share with the class.

Check your work

Your WBS decomposes to assignable work packages, satisfies the 100% rule, includes PM/testing/training effort, and every dictionary entry has testable acceptance criteria.

Activity 9 — Build the Schedule and Find the Critical Path with GenAI

Learning outcome: LO1 — Sequence activities, estimate durations and identify the critical path.

Goal: You will turn your work packages into a sequenced, estimated schedule, draw the network diagram, and compute the critical path — then check whether GenAI's arithmetic can be trusted.

What you'll produce

A Gantt chart, a schedule network diagram and a calculated critical path. (Tools: GenAI tool · Gantt Chart Template.xlsx · Schedule Network Diagram.xlsx.)

Activity package

Open activities/activity-09-build-the-schedule-and-find-the-critical-path-with-g/. It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Using the task data provided, propose a project schedule for building a CRM system. Preserve the stated dependencies, identify the critical path and show your forward-pass and backward-pass calculations.

Step-by-step

1. In your group, prompt GenAI: 'Generate a list of tasks with estimated durations for building a CRM system over 8 weeks.' Open 'Gantt Chart Template.xlsx'.
2. Enter the tasks and durations, then add dependencies. Classify each dependency as Mandatory, Discretionary, External or Internal — GenAI will not do this correctly for your context.
3. Apply three-point estimating to your five most uncertain tasks. Prompt: 'For each task give an optimistic, most likely and pessimistic duration.' Compute the PERT estimate yourself: $(O + 4M + P) / 6$.
4. Compare the PERT figure with GenAI's single-point estimate. Note how much optimism was hidden in the original number.
5. Open 'Schedule Network Diagram.xlsx'. Prompt: 'List 10 project tasks with durations and dependencies for developing an internal helpdesk ticketing system.'
6. Enter activities, durations and dependencies, then perform the forward and backward pass to find each activity's float.
7. Determine the critical path — the sequence with zero float — and calculate the total project duration.
8. Now verify GenAI: ask it to compute the critical path for the same network, and compare with your manual answer. Where they differ, work out which is right and why.
9. Share your critical path and total duration with the class.

Check your work

You can state the critical path, the total duration and the float on at least one non-critical activity, and you have checked GenAI's computed path against your own.

Activity 10 — Estimate Costs and Build the Budget with GenAI

Learning outcome: LO1/LO2 — Produce a defensible cost baseline with appropriate reserves.

Goal: You will apply the four estimating techniques to the same work, compare what each produces, and assemble a cost baseline with contingency and management reserves.

What you'll produce

A cost baseline built bottom-up from your WBS, with reserves and a documented estimating basis. (Tools: GenAI tool · Cost Benefit Analysis.xlsx · your WBS from Activity 8.)

Activity package

Open [activities/activity-10-estimate-costs-and-build-the-budget-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Estimate the cost of each work package in this WBS using bottom-up estimating. Show assumed rates and quantities separately from totals, then add contingency and management reserves without double counting.

Step-by-step

1. Using the WBS from Activity 8, prompt GenAI: 'Estimate the cost of each work package in this WBS using bottom-up estimating. Show your assumed rates and quantities separately from the totals.'
2. Ask for the same total three more ways: analogous ('estimate from a similar past project'), parametric ('use a rate per unit') and three-point ('give optimistic, most likely and pessimistic').
3. Compute the three-point estimate yourself using $C_e = (C_o + 4C_m + C_p) / 6$ and compare all four figures. Explain to your group why they differ and which you would defend to a sponsor.
4. Challenge the assumed rates. Prompt: 'Which of these rates did you assume rather than know? Mark each with [ASSUMPTION].' Replace them with rates from your own organisation.
5. Roll the work package costs up to a project total — this is your cost baseline before reserves.
6. Add a contingency reserve for the identified risks you will quantify in Topic 3, and a management reserve for unforeseen in-scope work. Prompt GenAI to suggest a percentage and to justify it, then decide your own figure.

7. Confirm you can state the difference: contingency reserve is inside the cost baseline for known risks; management reserve sits outside it for unknown work.
8. Share your baseline and the reserve percentages you chose, with your justification.

 Check your work

You have one cost baseline built bottom-up, four comparable estimates for the same work, no unmarked assumed rates, and a stated contingency and management reserve you can justify.

Topic 02 — Project Execution and Delivery with GenAI (LO2 · K3, A2)

Directing work · knowledge · quality · resources · team · communications

Key concepts

- Direct and manage project work — Turn the plan into deliverables; distinguish work performance data, information and reports.
- Manage project knowledge — Capture explicit knowledge and surface tacit knowledge; GenAI summarises lessons learned as you go.
- Quality management — Cost of quality, PDCA, DMAIC, Kaizen; fishbone, five whys and Pareto for root-cause analysis.
- Acquire and develop resources — Resource levelling vs smoothing; Tuckman's ladder; motivational theories.
- Manage the team — Conflict management approaches, emotional intelligence and interpersonal skills.
- Manage communications — The $N(N-1)/2$ channel formula, the Five Cs, push vs pull, and barriers to communication.
- Optimising resources and budget — Use GenAI to model resource and budget options, then justify the trade-off you choose.

Activity 11 — Direct Project Work and Report Performance with GenAI

Learning outcome: LO2 — Turn raw project data into work performance information and reports.

Goal: Executing a project generates a flood of raw data. You will use GenAI to turn that data into information, and information into a report a sponsor will actually read — while learning the distinction that PMBOK draws between the three.

What you'll produce

A project status report generated from raw work performance data. (Tools: GenAI tool · Project Manager Buddy GPT.)

Activity package

Open [activities/activity-11-direct-project-work-and-report-performance-with-gena/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Here is raw work performance data from my project [paste]. Compare it with the plan, distinguish data from interpretation, and produce a one-page status report for an executive sponsor.

Step-by-step

1. In your group, write out ten lines of raw work performance data for a project in progress — for example '14 defects open', 'actual cost to date \$62,000', '3 change requests raised'.
2. Confirm the distinction before you prompt: Data is the raw observation; Information compares two or more data points to make it meaningful; a Report is the representation intended to raise action or awareness.
3. Prompt GenAI: 'Here is raw work performance data from my project [paste]. Convert it into work performance information by comparing against the plan, then produce a one-page status report for an executive sponsor.'
4. Check what it inferred. It will assume a baseline you never gave it — identify every comparison it invented and supply the real planned figure.
5. Prompt it to sharpen the report: 'Rewrite this so the first three lines tell the sponsor whether the project is on track, what the single biggest problem is, and what decision I need from them.'
6. Add a version control block — version number, date/time stamp and author — as required for project document control.
7. Use the Persona pattern to test it: 'Act as a sceptical project sponsor. What would you challenge in this report?' Address the challenges it raises.
8. Share your report and state which figure GenAI fabricated before you corrected it.

Check your work

Your report distinguishes data from information, every comparison uses a real baseline figure, and it opens with status, biggest problem and the decision required.

Activity 12 — Manage Project Knowledge with GenAI

Learning outcome: LO2 — Capture explicit and tacit knowledge so it is reusable by the organisation.

Goal: Knowledge that lives only in people's heads leaves when they do. You will use GenAI to turn messy project conversations into a structured, searchable knowledge asset and a lessons-learned register.

What you'll produce

A lessons-learned register and a knowledge asset entry for your project. (Tools: GenAI tool · Claude (long-document handling).)

Activity package

Open [activities/activity-12-manage-project-knowledge-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

From these project records [paste], extract decisions and owners, open questions, lessons learned, and tacit knowledge that is implied but not stated. Quote the evidence used for each extraction.

Step-by-step

1. In your group, write a realistic 'messy input': meeting notes, a chat thread and an email chain about a project problem — roughly a page in total. Invent it if you need to.
2. Prompt GenAI: 'From these project records, extract: (1) decisions made and who made them, (2) open questions, (3) lessons learned, (4) tacit knowledge that is implied but never stated explicitly.'
3. Examine point 4 closely — surfacing tacit knowledge (the 'everyone knows that the vendor is slow in December' type of insight) is where GenAI adds real value here.
4. Build a lessons-learned register entry for each lesson: what happened, the impact, the root cause and the recommendation for future projects.
5. Prompt: 'Rewrite each lesson so it is actionable by a project manager who knows nothing about this project.' Vague lessons are useless lessons.
6. Decide what belongs in the organisational process assets versus what is project-specific, and mark each entry accordingly.
7. Discuss as a group: what are the confidentiality risks of pasting real project records into a public GenAI tool, and what would your organisation's policy allow?

Check your work

Your register has actionable lessons with root causes, at least one item of genuinely tacit knowledge surfaced, and your group can state your organisation's data-confidentiality limits.

Activity 13 — Root Cause Analysis with GenAI — Fishbone, Five Whys and Pareto

Learning outcome: LO2 — Diagnose the root cause of a delivery problem, not just its symptoms.

Goal: When delivery goes wrong, the visible problem is rarely the real one. You will run three complementary root-cause techniques on the same problem with GenAI's help and compare what each reveals.

What you'll produce

A fishbone diagram, a five-whys chain and a Pareto analysis of the same delivery problem. (Tools: GenAI tool · Fishbone Diagram Template.pptx · Pareto Principle Template.xlsx.)

Activity package

Open [activities/activity-13-root-cause-analysis-with-genai-fishbone-five-whys-an/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Analyse the issue records provided and identify the most likely root causes of recurring project deadline misses. Produce a fishbone structure, a five-whys chain and a Pareto-ready category table without inventing evidence.

Step-by-step

1. In your group, choose one real delivery problem — recurring missed deadlines, rework, defects escaping to production.
2. Open 'Fishbone Diagram Template.pptx'. Prompt GenAI: 'What are the possible root causes of recurring project deadline misses in IT infrastructure upgrades?' — substituting your problem.
3. Sort the suggested causes into the fishbone categories (People, Process, Technology, Environment, Materials, Measurement). Reclassify anything the AI put in the wrong bone.
4. Add the causes GenAI could not know — the ones specific to your organisation's politics, tooling or history. These are usually the real ones.
5. Now run Five Whys on the single most likely cause. Prompt: 'Apply the Five Whys method to [problem] to uncover the root cause and identify potential solutions.'
6. Critique the chain. GenAI tends to produce a plausible but shallow chain that stops at 'insufficient planning'. Push each 'why' until you reach something you could actually change.

7. Open 'Pareto Principle Template.xlsx'. Prompt GenAI to generate a frequency count of defect or delay categories, enter them, and identify the vital few causing roughly 80% of the impact.
8. Compare the three techniques: which surfaced the actionable cause? Share your conclusion with the class.

 Check your work

You have all three artifacts for one problem, you added at least two organisation-specific causes the AI missed, and your five-whys chain ends at something you can change.

Activity 14 — Continuous Improvement with GenAI — PDCA and DMAIC

Learning outcome: LO2 — Structure an improvement cycle and control the result.

Goal: Having found a root cause, you need a disciplined way to fix it and make the fix stick. You will run the same improvement through both PDCA and DMAIC and see where they differ.

What you'll produce

A completed PDCA cycle and a DMAIC project definition for your improvement. (Tools: GenAI tool · Plan Do Check Act.pptx · DMAIC.pptx.)

Activity package

Open [activities/activity-14-continuous-improvement-with-genai-pdca-and-dmaic/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Design a PDCA cycle to address [root cause]. For Plan, state the evidence and prioritised solution; for Do, define a small pilot; for Check, define the success measure; for Act, state what happens if the pilot succeeds or fails.

Step-by-step

1. Take the root cause you identified in Activity 13. Open 'Plan Do Check Act.pptx'.
2. Prompt GenAI: 'Design a PDCA cycle to address [root cause]. For Plan, state the data reviewed and the prioritised solutions; for Do, define a small pilot; for Check, define the measure of success; for Act, state what happens if it works and if it does not.'
3. Sharpen the Check stage — this is where AI drafts are weakest. Insist on a specific number and a date: 'rework hours per sprint below 8 by 30 September', not 'improved efficiency'.

4. Complete the PDCA template. Confirm your Do stage is genuinely a small pilot rather than a full rollout.
5. Now open 'DMAIC.pptx' and prompt: 'Structure the same improvement as a DMAIC project. Define the gap, state how I would measure the current problem with real data, analyse root cause, propose the improvement, and specify the control that keeps the process in place.'
6. Compare the two: PDCA is a fast learning loop; DMAIC is a heavier, data-driven method for a chronic problem. State which fits your improvement and why.
7. Pay attention to the Control stage — ask GenAI 'what would cause this improvement to decay after six months?' and build a control against it.
8. Share your improvement plan with the class.

 Check your work

Your Check/Control stages specify a number and a date, your Do stage is a pilot, and you can justify choosing PDCA or DMAIC for this problem.

Activity 15 — Optimise Resources and Budget with GenAI

Learning outcome: LO2 — Resolve over-allocation and budget pressure by modelling options with GenAI (K3).

Goal: This is the heart of LO2. A resource is over-allocated and the budget is under pressure — you will use GenAI to generate genuine alternatives, evaluate the trade-offs, and defend the option you choose.

What you'll produce

A resource optimisation decision with a documented trade-off analysis. (Tools: GenAI tool · your schedule from Activity 9 · Gantt Chart Template.xlsx.)

Activity package

Open [activities/activity-15-optimise-resources-and-budget-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

I have an over-allocated resource on these activities [paste] and a 15% budget reduction. Compare resource levelling, resource smoothing, crashing and fast tracking, including cost, schedule, quality and risk trade-offs.

Step-by-step

1. Take your schedule from Activity 9 and deliberately create a problem: assign one person to two activities running in the same period, and cut the budget by 15%.

2. Confirm the two techniques before prompting: resource LEVELLING adjusts the schedule to fix over-allocation and may move the critical path; resource SMOOTHING adjusts only within available float, so the critical path does not change.
3. Prompt with the Alternative Approaches pattern: 'I have an over-allocated resource on these activities [paste] and a 15% budget cut. Suggest at least three options — including resource levelling, resource smoothing, crashing and fast tracking. Compare pros, cons, cost impact and schedule impact in a table.'
4. Check each option against the definitions. GenAI frequently confuses levelling with smoothing and describes crashing without noting it only works on the critical path.
5. For any crashing option, verify it targets a critical path activity — crashing a non-critical activity buys you nothing. For fast tracking, identify the rework risk it creates.
6. Apply the Cognitive Verifier pattern: 'Break down which option gives the best schedule outcome for the least cost increase and the least added risk, then combine into a recommendation.'
7. Make your group's decision, adjust the Gantt chart accordingly, and write three sentences justifying it to a sponsor.
8. Present your recommendation and state one thing GenAI got wrong about levelling, smoothing, crashing or fast tracking.

 Check your work

Your chosen option is correctly named, crashing (if used) targets the critical path, and you can justify the cost/schedule/risk trade-off in three sentences.

Activity 16 — Develop and Manage the Team with GenAI

Learning outcome: LO2 — Advise staff in resolving challenges during project delivery (A2).

Goal: Delivery problems are usually people problems. You will use GenAI to prepare for a difficult conversation and to select a conflict management approach — while judging where AI advice on people is genuinely useful and where it is not.

What you'll produce

A prepared difficult conversation and a team development plan mapped to Tuckman's stages. (Tools: GenAI tool · Claude · Scrum Agile Transformation Studio (alfredang.github.io/scrum).)

Activity package

Open [activities/activity-16-develop-and-manage-the-team-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

My project team is in the [stage] stage of Tuckman's model and showing these behaviours [paste]. Recommend leadership actions that can move the team forward, and explain which observations support each action.

Step-by-step

1. In your group, pick a realistic delivery challenge: a team member missing commitments, two members in open conflict, or a team stuck in the storming stage.
2. Locate your team on Tuckman's ladder — Forming, Storming, Norming, Performing or Adjourning — and prompt GenAI: 'My project team is in the [stage] stage and showing [behaviours]. What leadership actions move them to the next stage?'
3. Now prepare a difficult conversation using the Reflection pattern: 'I need to give feedback to a team member who [situation]. They are talented and I want this to help their development. Explain the reasoning and assumptions behind your recommendation for approaching this meeting.'
4. Evaluate the advice critically. It will be reasonable but generic — it knows nothing about this person, your relationship, or your culture. Note precisely which parts you would not use.
5. Review the five conflict management approaches — Smooth/Accommodate, Withdraw/Avoid, Compromise/Reconcile, Force/Direct, Collaborate/Problem Solve — and prompt GenAI to recommend one for your situation with its reasoning.
6. Challenge it: 'When would Force/Direct be the right choice here?' A good PM knows collaboration is not always correct — safety and deadline-critical decisions sometimes require directing.
7. Open <https://alfredang.github.io/scrum/> — one member clicks 'Create Workspace', selects a role (Product Owner, Scrum Master, Developer, Stakeholder or Agile Coach) and shares the workspace ID; everyone joins with their own role.
8. Run a retrospective in the studio on your delivery challenge, using GenAI to suggest retrospective prompts. Use the Export panel to download the workspace as CSV or a printable HTML report as your evidence.
9. Discuss as a group: what should a PM never delegate to GenAI when it concerns a specific person?

Check your work

You have a conversation plan you would actually use, a named conflict approach with justification, an exported retrospective file, and a clear line on what not to delegate to AI.

Activity 17 — Plan and Manage Communications with GenAI

Learning outcome: LO2 — Build a communications plan that matches each stakeholder's real needs.

Goal: You will calculate how communication complexity grows, then use GenAI to build a communications plan and adapt a single message for very different audiences.

What you'll produce

A communications management plan and the same update written for three audiences.
(Tools: GenAI tool · your stakeholder register from Activity 4.)

Activity package

Open [activities/activity-17-plan-and-manage-communications-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Build a communications management plan from the stakeholder data provided. For each stakeholder specify information needed, purpose, format, level of detail, frequency, channel and sender.

Step-by-step

1. Calculate the communication channels in your project team using $N \times (N - 1) / 2$. For a team of 10 that is 45. Recalculate for 15 people and note how fast it grows.
2. Using your stakeholder register from Activity 4, prompt GenAI: 'Build a communications management plan. For each stakeholder specify: information needed, reason, format, level of detail, frequency, channel, and sender.'
3. Check for the failure mode: the AI will propose sending everything to everyone weekly. Cut it back to what each stakeholder genuinely needs, and use push vs pull deliberately — push for what must be read, pull (a dashboard or portal) for reference information.
4. Now write one piece of bad news — a two-week slip. Prompt GenAI to write it three ways: for the executive sponsor, for the delivery team, and for an external customer.
5. Compare the three against the Five Cs: Correct, Concise, Clear, Coherent, Controlled. Rewrite whichever version fails.
6. Check the sponsor version leads with the impact and the decision required, not with an explanation of what happened.
7. Identify which communication barriers apply in your project — language, cultural, physical, psychological, perceptual, information overload, jargon — and add one mitigation for each to your plan.
8. Share your three versions and explain what you changed between them and why.

Check your work

Your plan gives each stakeholder a specific frequency and channel, uses push and pull deliberately, and your three message versions genuinely differ in what they lead with.

Topic 03 — Managing Project Risks with GenAI (LO3 · K1, A3, A4)

Risk · procurement · monitoring & control · earned value · closure

Key concepts

- Risk vs issue — Risks are future and may be positive or negative (risk register); issues are present and always negative (issue log).
- Identify and analyse risk — RBS, brainstorming, SWOT and the Delphi technique; qualitative probability/impact then quantitative analysis.
- Quantitative analysis — Sensitivity (tornado) charts and decision-tree expected monetary value.
- Plan risk responses — Avoid, transfer, mitigate, accept — escalate, exploit, share, enhance; secondary and residual risk.
- Procurement — RFI/RFP/RFQ, contract types, claims and alternative dispute resolution.
- Monitor and control — Integrated change control, validate and control scope, schedule and cost.
- Earned value analysis — PV, EV, AC; CV, SV, CPI, SPI; EAC and ETC forecasting.
- Control quality & close — Check sheets, control charts and the rule of seven; then close the project and capture lessons learned.

Activity 18 — Identify Project Risks with GenAI

Learning outcome: LO3 — Anticipate potential risks to project delivery (A3).

Goal: GenAI is genuinely excellent at risk identification — it will suggest risks your team would never think of. It is correspondingly poor at knowing which of them matter to you. You will use both facts.

What you'll produce

A populated risk register with risks sourced from GenAI and from your team. (Tools: GenAI tool · Project Manager Buddy GPT · Risk Register template.)

Activity package

Open [activities/activity-18-identify-project-risks-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Act as an expert project manager in the [industry] working on a [project type]. Use the supplied project facts to draft a risk register covering

cause, event, effect, category, owner and evidence. Separate facts from assumptions.

Step-by-step

1. In your group, spend five minutes brainstorming risks to your project WITHOUT any AI. Write them down — this is your baseline and you will compare against it.
2. Now prompt GenAI with the Risk Matrix Template pattern: 'Imagine you are an expert project manager in the [Industry], focusing on [Project type] projects. Develop a comprehensive risk matrix. For each risk include Risk name, Description, Level of impact (high/medium/low) and Likelihood (high/medium/low). Cover budget, schedule, technology, regulatory, resource, scope creep, stakeholder, data security, quality and external dependency categories. Output as a table.'
3. Compare against your manual list. Count how many risks the AI found that your team missed — typically it will be more than half.
4. Now find the opposite: which risks did YOUR team identify that the AI missed? These are almost always the organisation-specific ones, and they are usually the ones that actually materialise.
5. Use the Risk Breakdown Structure as a checklist to find further gaps — work down from high-level categories (External, Organisational, Technical, Project Management) to specific risks.
6. For each risk, confirm it is written as a proper risk statement: cause → risk event → effect. 'The vendor may be late' is not enough; 'Because the vendor has one integration engineer, their resource may be unavailable in December, delaying UAT by two weeks' is.
7. Distinguish risks from issues as you go: risks are future and may be positive or negative and go in the risk register; issues are present, always negative, and go in the issue log.
8. Assign an owner to every risk and share your register with the class.

✓ Check your work

Your register has risks from both sources, every entry is written as cause → event → effect with an owner, and you can name one risk your team found that the AI missed.

Activity 19 — Qualitative Risk Analysis with GenAI

Learning outcome: LO3 — Prioritise risks by probability and impact for further action.

Goal: You cannot act on fifty risks. Qualitative analysis prioritises them so you spend your attention where it matters — and you will test how consistent GenAI's scoring actually is.

What you'll produce

A scored probability/impact matrix with a prioritised risk shortlist. (Tools: GenAI tool · Probability and Impact Matrix template.)

Activity package

Open [activities/activity-19-qualitative-risk-analysis-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Score each supplied risk for probability and impact from 1 to 5, calculate the risk score, place it in a probability-impact matrix and explain any score that is not directly supported by the data.

Step-by-step

1. Take your risk register from Activity 18. Prompt GenAI: 'Score each risk for probability (1-5) and impact (1-5), calculate the risk score, and place each in a probability/impact matrix.'
2. Run the same prompt a second time in a new chat. Compare the two sets of scores — you will find they differ. Discuss what that inconsistency means for relying on AI scoring.
3. Agree your group's own scores for the top ten risks. Use GenAI's output as a prompt for discussion, not as the answer.
4. Plot the risks on the probability/impact matrix and identify the red zone — high probability, high impact — that needs a response plan.
5. Apply the additional risk factors to your top five: urgency, proximity, dormancy, manageability, controllability, strategic impact, connectivity and propinquity.
6. Prompt: 'For these top five risks, assess proximity (how soon it could occur) and dormancy (how long before we would notice it).' A high-dormancy risk needs a detection mechanism, not just a response.
7. Update your risk register with the agreed scores, the factor assessments and a priority ranking. Share your top three risks with the class.

Check your work

Your top ten risks have group-agreed scores, your top five carry proximity and dormancy assessments, and you can explain why AI scoring alone is not sufficient.

Activity 20 — Quantitative Risk Analysis with GenAI — Tornado Charts and Decision Trees

Learning outcome: LO3 — Quantify risk exposure to support a financial decision.

Goal: When a decision carries real money, qualitative scoring is not enough. You will run a sensitivity analysis and a decision tree, using GenAI for the setup and doing the arithmetic yourself.

What you'll produce

A tornado chart identifying your most sensitive risk variables and a decision tree with expected monetary values. (Tools: GenAI tool · Tornado Chart - Sensitivity Analysis.xlsx · Decision Tree Analysis.xlsx.)

Activity package

Open [activities/activity-20-quantitative-risk-analysis-with-genai-tornado-charts/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Use the supplied high, base and low values to rank the ten variables by their effect on project cost or schedule. Produce tornado-chart data and one decision tree with expected monetary values, showing every calculation.

Step-by-step

1. In your group, open 'Tornado Chart - Sensitivity Analysis.xlsx'.
2. Prompt GenAI: 'List 10 risk variables that could affect the cost or schedule of a digital transformation project. Estimate high and low values for each.'
3. Enter each variable with its high/low range and plot the bars, longest at the top. The variables at the top of the tornado are where your management attention belongs.
4. Sanity-check the ranges. GenAI produces symmetric ranges by default; real project risks are usually asymmetric — things go wrong further than they go right. Widen the downside where your experience says so.
5. Open 'Decision Tree Analysis.xlsx'. Define a real either/or decision from your project — build vs buy, or two vendors.
6. Prompt GenAI: 'Structure this decision as a decision tree. For each branch give the cost, the possible outcomes with their probabilities, and the payoff of each outcome.'
7. Calculate the Expected Monetary Value for each branch yourself: $EMV = \text{sum of (probability} \times \text{payoff) for each outcome, minus the branch cost}$. Do NOT accept the AI's arithmetic — verify every multiplication.
8. Compare your computed EMV with GenAI's. Where they differ, find the error. This is the single most important habit when using AI for quantitative work.
9. State your recommended decision and the EMV that supports it. Share with the class.

Check your work

Your tornado chart is ordered by sensitivity with justified asymmetric ranges, and you have

hand-verified the EMV of every decision tree branch.

Activity 21 — Plan Risk Responses with GenAI

Learning outcome: LO3 — Devise risk mitigation measures to ensure project success (A4).

Goal: A risk register with no responses is a list of things you will be surprised by. You will plan a response for every priority risk and check for the second-order effects that responses create.

What you'll produce

A response plan for every priority risk, with owners, triggers and fallback plans. (Tools: GenAI tool · your risk register from Activities 18-19.)

Activity package

Open [activities/activity-21-plan-risk-responses-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

For each priority risk [paste], recommend one primary response strategy from avoid, transfer, mitigate, accept, escalate, exploit, share or enhance. Give the action, owner, trigger, fallback and response cost.

Step-by-step

1. Confirm the eight response strategies. For threats: Avoid, Transfer, Mitigate, Accept. For opportunities: Exploit, Share, Enhance, Accept. Escalate applies to both when the risk is outside your authority.
2. Prompt GenAI: 'For each of these priority risks [paste], recommend a response strategy from avoid, transfer, mitigate, accept, escalate, exploit, share or enhance. Give the specific action, an owner role, the trigger condition, and the estimated cost of the response.'
3. Audit the strategy names. GenAI routinely labels a mitigation as an avoidance — avoiding a risk means eliminating the cause entirely, usually by changing the plan, not by reducing the likelihood.
4. Check the cost logic: a response costing more than the risk exposure it removes is a bad response. Prompt: 'For each response, compare its cost against the risk exposure (probability x impact). Flag any response that costs more than the exposure.'

5. Identify SECONDARY risks — new risks created by your responses. Prompt: 'What new risks does each of these responses introduce?' Transferring risk to a vendor, for instance, creates vendor dependency risk.
6. Identify RESIDUAL risk — what remains after the response is implemented. Every accepted residual risk needs a contingency plan and a contingency reserve.
7. Write a fallback plan for your top three risks: what you do if the primary response fails.
8. Define the trigger condition for each response — the observable signal that tells you to act. A response with no trigger is never implemented in time.
9. Update your register and share your top three responses with their triggers and fallbacks.

 Check your work

Every priority risk has a correctly-named strategy, an owner, an observable trigger, an identified secondary and residual risk, and your top three have fallback plans.

Activity 22 — Plan Procurement and Manage Contracts with GenAI

Learning outcome: LO3 — Select the right contract type and manage procurement risk.

Goal: Procurement transfers risk — but only if you choose the right contract type. You will use GenAI to draft procurement documents and to reason about where risk actually sits in each contract type.

What you'll produce

A procurement strategy with a justified contract type and a drafted RFP outline. (Tools: GenAI tool · Claude.)

Activity package

Open [activities/activity-22-plan-procurement-and-manage-contracts-with-genai/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Compare Firm Fixed Price, Cost Plus Fixed Fee and Time and Materials for this procurement. State who carries cost-overrun risk, when each contract fits, and recommend one option using the supplied evaluation criteria.

Step-by-step

1. In your group, identify something your project must procure externally — a vendor, consultant, software or equipment.

2. Decide which bid document fits: RFI when you need information on what sellers can provide; RFP when the solution is not obvious and you need sellers to propose one; RFQ when you know exactly what you want and need a price.
3. Prompt GenAI: 'Draft an outline [RFI/RFP/RFQ] for [what you are procuring]. Include the scope of work, deliverables, evaluation criteria with weightings, and the submission requirements.'
4. Now the important question. Prompt: 'Compare Firm Fixed Price, Cost Plus Fixed Fee and Time and Materials for this procurement. For each, state who carries the risk of a cost overrun and under what circumstances it is the right choice.'
5. Verify the risk logic yourself: fixed price puts overrun risk on the SELLER and suits well-defined scope; cost reimbursable puts it on the BUYER and suits changing work; T&M suits an undefined statement of work and staff augmentation.
6. Choose your contract type and write two sentences justifying it in terms of who should carry the risk given how well you know the scope.
7. Prompt GenAI to list the contract components you must include: description of work, delivery schedule, responsibilities of both parties, price and payment terms, termination provisions, guarantees and warranties, intellectual property, and security/confidentiality/data privacy.
8. Add a claims and dispute clause. Prompt: 'What should the claims process and alternative dispute resolution clause say?' Confirm you understand a claim arises when buyer and seller cannot agree that a change occurred.
9. Share your contract type choice and your risk justification with the class.

 Check your work

You have the right bid document type, a contract type justified by who should carry scope risk, and a component checklist including IP, data privacy and dispute resolution.

Activity 23 — Earned Value Analysis with GenAI

Learning outcome: LO3 — Measure project performance objectively and forecast the outcome.

Goal: Earned value is the one technique that tells you objectively whether you are ahead or behind, over or under. You will compute it by hand first, then test whether GenAI gets it right.

What you'll produce

A complete earned value analysis with variances, indices and a forecast. (Tools: GenAI tool · calculator.)

Activity package

Open activities/activity-23-earned-value-analysis-with-genai/. It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Using the supplied baseline, completion and actual-cost data, calculate PV, EV, AC, CV, SV, CPI, SPI, EAC and ETC. Show the formula and substitution for every result, then interpret the project status.

Step-by-step

1. Work this classic scenario by hand, individually. Task: develop and install ten printer drivers. Budget: \$100,000 (\$10K per driver). Time: 10 weeks (1 driver per week). At week 5: 4 drivers are complete and \$47,500 has been spent.
2. Compute the three base values. PV = the budget for work SCHEDULED by now. EV = the budget for work ACTUALLY DONE. AC = what has actually been spent.
3. Confirm your answers: PV = \$50,000 (5 drivers scheduled x \$10K). EV = \$40,000 (4 drivers done x \$10K). AC = \$47,500.
4. Compute the variances. Cost Variance CV = EV - AC = \$40,000 - \$47,500 = -\$7,500 (negative is bad — over budget). Schedule Variance SV = EV - PV = \$40,000 - \$50,000 = -\$10,000 (negative is bad — behind schedule).
5. Compute the indices. CPI = EV / AC = 40,000 / 47,500 = 0.84. SPI = EV / PV = 40,000 / 50,000 = 0.80. Both below 1 — over budget and behind schedule.
6. Forecast the outcome. EAC = BAC / CPI = 100,000 / 0.84 = approximately \$119,000. ETC = EAC - AC. State what you would tell the sponsor.
7. NOW test GenAI. Give it the same scenario and ask for PV, EV, AC, CV, SV, CPI, SPI and EAC. Compare every figure against yours.
8. Where it differs, identify the error. GenAI commonly confuses EV with AC, or computes EAC without using CPI. This is precisely why you must be able to do this by hand.
9. Discuss as a group: the assessment is open book, but the arithmetic is yours. What is the minimum you must be able to compute without help?

✓ Check your work

You can compute PV, EV, AC, CV, SV, CPI, SPI and EAC unaided for a fresh scenario, and you found at least one error in the AI's version.

Activity 24 — Control Quality with GenAI — Check Sheets and Control Charts

Learning outcome: LO3 — Detect whether a process is stable and in control.

Goal: Quality control is about detecting problems from data rather than opinion. You will collect data on a check sheet and plot a control chart, applying the rule of seven.

What you'll produce

A check sheet of collected data and a control chart with limits and a trend assessment.
(Tools: GenAI tool · Check Sheets.xlsx · Control Chart.xlsx.)

Activity package

Open [activities/activity-24-control-quality-with-genai-check-sheets-and-control/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Use the supplied defect log to produce a check-sheet summary and control-chart data. Calculate the centre line and control limits, identify any special-cause signal and explain what additional evidence is needed.

Step-by-step

1. In your group, open 'Check Sheets.xlsx'. Prompt GenAI: 'Design a check sheet to collect defect data for [your project's deliverable]. Suggest the defect categories and the collection frequency.'
2. Confirm the categories are mutually exclusive — a defect must fall into exactly one. Fix any overlap the AI created.
3. Populate the check sheet with 20 data points, either real data or a plausible set your group agrees on.
4. Open 'Control Chart.xlsx'. Enter your data and compute the mean, the upper control limit and the lower control limit.
5. Understand the distinction: control limits come from the process itself (what it actually does); specification limits come from the customer (what they require). A process can be in control and still fail the specification.
6. Apply the rule of seven: seven consecutive points on the same side of the mean, or seven consecutively increasing or decreasing, indicates a trend requiring investigation — even if no point breaches a control limit.
7. Prompt GenAI: 'Analyse this control chart data. Is the process in control? Are there any runs of seven? What would you investigate?' Verify its answer against your own reading of the chart.
8. Follow the full control quality sequence: check sheet → histogram → Pareto → cause-and-effect → scatter analysis. State which step you would do next and why.
9. Share whether your process is in control and what you would investigate.

✓ Check your work

Your control chart has a mean and both control limits, you have checked for runs of seven, and you can distinguish control limits from specification limits.

Activity 25 — Monitor, Control Change and Close the Project with GenAI

Learning outcome: LO3 — Control change through the project and close it properly.

Goal: The final activity runs the end of the project lifecycle: processing a change request through integrated change control, then closing the project and capturing lessons learned with GenAI's help.

What you'll produce

A processed change request with impact analysis, and a project closure report. (Tools: GenAI tool · Digital AI Transformation Roadmap Studio (alfredang.github.io/digitaltransformation/) · Change Request template.)

Activity package

Open [activities/activity-25-monitor-control-change-and-close-the-project-with-ge/](#). It contains the Markdown activity guide and PDF rendition, a prompt sheet, a detailed instruction manual and a formatted mock-data workbook.

Core prompt

Analyse each change request for its effect on scope, schedule, cost, quality, resources and risk. State confidence in each estimate, recommend an approve, reject or defer decision, and draft the closure actions for approved changes.

Step-by-step

1. In your group, raise a realistic change request against your project — a new regulatory requirement, or a stakeholder asking for an extra feature.
2. Classify it correctly: is it corrective action, preventive action, defect repair, or a change to a baseline? Only the last requires a baseline update.
3. Prompt GenAI: 'Analyse the impact of this change request on scope, schedule, cost, quality and risk. For each, state the impact and how confident you are in the estimate.'
4. Check the impacts it forgot. It will usually cover scope, schedule and cost, and skip the knock-on effects on quality, risk, resources and stakeholder expectations.
5. Run it through integrated change control: raise the request, analyse the impacts, record it in the change log, take it to the Change Control Board for approval/deferral/rejection, and communicate the outcome back to the requester.
6. Now open <https://alfredang.github.io/digitaltransformation/> — one member enters their name and clicks 'Create New Roadmap'; the others join with the Roadmap ID or QR code from the Invite button.
7. Add your project's remaining initiatives (title, phase, status, priority, owner, impact 1-10, effort 1-10, start month, duration). Prompt GenAI to propose the impact and effort scores, then adjust them as a group.

8. Read the Impact vs Effort matrix. Identify what falls in Quick Wins versus Thankless, and add your top risks to the Risk Register tab with likelihood, impact and mitigation. Add two or three KPIs with current and target values.
9. Move to closure. Prompt GenAI: 'Draft a project closure report covering: acceptance of deliverables, transition and handover requirements, contract closure, lessons learned on scope changes, schedule impacts, risks and issues, stakeholder and vendor relationships, and recommendations.'
10. Complete the closure checklist: deliverables accepted against acceptance criteria, product transitioned with training and hypercare, contracts and accounts closed, project information archived to organisational process assets, resources released.
11. Use the Export button in the Roadmap Studio to capture your roadmap, and share your closure report with the class.

 Check your work

Your change request is correctly classified with impacts on all five dimensions, it has been through the CCB path, and your closure report covers acceptance, transition, contracts, archiving and resource release.

Applying This at Work

Start with one artifact

Do not try to change everything at once. Pick a single artifact you produce regularly — a status report, a risk register or a set of meeting minutes — and draft the next one with GenAI. Compare how long it took against your usual approach, and note what you had to correct.

Reuse your prompt library

The prompt library you built in Activity 1 is the most portable thing you take from this course. Every prompt in it has already been tested on a project you understand. Keep adding to it, and record the critique alongside each prompt so you remember what the model tends to get wrong.

The five checks that catch most GenAI errors

- Arithmetic — recompute every number yourself. Earned value, EMV, PERT and critical path are all commonly wrong.
- Sources — if it cites a source, open it. Invented citations are common and look entirely plausible.
- Assumptions — ask it to mark every figure it assumed rather than knew, then replace those figures.
- Completeness — ask it what it left out. It reliably omits project management effort, testing, training and handover.

- Context — it knows nothing about your organisation's politics, history or constraints. That gap is where you add value.

Where GenAI should not go

- Decisions about a specific named person — performance, discipline or selection.
- Anything involving client data, personal data or commercially sensitive figures in a public tool.
- Final approval of a baseline, a budget or a contract — those remain human decisions.
- Any output you would not be willing to defend as your own work.

Consolidate Your Learning

Use this checklist after the course: work through each point with the slides, this Learner Guide and your own activity outputs.

- Be able to compute earned value unaided: PV, EV, AC, CV, SV, CPI, SPI, EAC and ETC (see Activity 23).
- Be able to name and distinguish the eight risk response strategies and give an example of each (Activity 21).
- Be able to explain the difference between resource levelling and smoothing, and between crashing and fast tracking (Activity 15).
- Be able to state what a Project Charter must contain, and what a WBS dictionary entry holds (Activities 3 and 8).
- Be able to describe at least four prompting patterns and when you would use each.

Glossary

- **Generative AI (GenAI)** — Tools that create new text, images, code or ideas from a prompt by predicting likely output from patterns in training data.
- **Prompt** — The instruction you give a GenAI tool. A structured prompt states role, context, task, format and how to flag assumptions.
- **Hallucination** — Fluent, confident output that is factually wrong — most often an invented figure, source or citation.
- **Project Charter** — The document that authorises the project, names the sponsor and PM, and fixes objectives, budget and success criteria.
- **Business Case** — A documented economic feasibility study establishing the benefits that justify the project work.
- **Stakeholder** — Anyone affected by the project, or who believes they are affected.
- **SEAM** — Stakeholder Engagement Assessment Matrix — records each stakeholder's Current and Desired engagement level.
- **RACI** — A responsibility assignment matrix: Responsible, Accountable, Consulted, Informed. Exactly one Accountable per activity.

- **RTM** — Requirements Traceability Matrix — traces each requirement from its source through to its test case and deliverable.
- **WBS** — Work Breakdown Structure — the decomposition of project scope into assignable work packages.
- **Work Package** — A piece of work small enough to be assigned to one person or team.
- **Critical Path** — The sequence of activities with zero float that determines the shortest possible project duration.
- **Float** — Free float is how long an activity can slip without delaying its successor; total float without delaying the project.
- **PERT** — A three-point estimate weighting the most likely value: $(O + 4M + P) \div 6$.
- **Cost Baseline** — The approved time-phased budget: work package costs plus the contingency reserve.
- **Contingency Reserve** — Budget held inside the cost baseline for identified risks. Controlled by the project manager.
- **Management Reserve** — Budget held outside the cost baseline for unforeseen in-scope work. Requires management approval.
- **Risk vs Issue** — A risk is a future event that may be positive or negative (risk register); an issue is a present, always-negative problem (issue log).
- **Secondary Risk** — A new risk created directly by implementing a risk response.
- **Residual Risk** — The risk that remains after a response has been implemented, and must be knowingly accepted.
- **Trigger Condition** — The observable signal that a risk is developing and the response should be implemented.
- **EMV** — Expected Monetary Value — the sum of (probability $\times$ payoff) across a decision branch's outcomes.
- **Earned Value (EV)** — The budget for the work actually completed, expressed in budget terms.
- **CPI / SPI** — Cost and Schedule Performance Index. $EV \div AC$ and $EV \div PV$. Above 1 is good; below 1 is bad.
- **EAC / ETC** — Estimate At Completion ($BAC \div CPI$) and Estimate To Complete ($EAC - AC$).
- **Rule of Seven** — On a control chart, seven consecutive points on one side of the mean — or seven rising or falling — indicates a trend to investigate.
- **Integrated Change Control** — The process of raising, impact-analysing, logging, approving and communicating a change to a baseline.
- **Scope Creep** — Unapproved scope added outside change control. Gold plating is delivering extras the customer never asked for.